ACSAR-E

Autonomous Cognitive Situational Awareness Robot

A Policy-Constrained Decision Intelligence Framework for Explainable Quadruped Autonomy

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1 PROBLEM SCOPE

Problem Statement: Current patrol robots can navigate and detect objects, but they cannot justify their actions or prove they are following institutional policy. To meet EU AI Act and PDPA 2022 standards, enterprises now require on-device transparency and human oversight that functions without cloud dependency.

In plain terms: today's robots act but cannot explain themselves. Regulators now require a clear paper trail for every robot decision like a logged incident report generated on the robot itself, with no internet connection required.

2 RESEARCH QUESTIONS

- **RQ1: Performance Safety Compliance** Does the proposed PGDS significantly reduce critical policy violations compared to Rule-only and VLM-only baselines when deployed on edge hardware (Orin NX)? Metric: Macro-avg F_1 (weighted for False Negatives) across 200 labeled scenarios.
- **RQ2: Human-in-the-Loop Transparency** Are the natural language justifications generated by PGDS accurate and accessible to non-technical operators? Metric: System Usability Scale (SUS) ≥ 68; $\geq 80\%$ comprehension rate; and high Justification-Policy fidelity.

3 LITERATURE POSITIONING

Current VLM planners and safety frameworks lack the deterministic rigor required for institutional compliance. The table below positions ACSAR-E against the current state-of-the-art:

Table 1: Comparison of Safety and Planning Frameworks

Approach	Representative Models	Enforcement Type	Auditability
VLM Planning	SayPlan, Code-as-Policies	None (Probabilistic)	None
Safety Alignment	Constitutional AI, RLHF	Soft (Prompt-based)	Opaque
ACSAR-E	Proposed System	Hard (Rule-layer)	Clause-to-Action

The Research Gap: While prior works focus on functional planning, they fail in regulated environments. ACSAR-E addresses this by replacing probabilistic "soft prompts" with an executable rule layer, providing *deterministic policy-constrained execution with traceability* at the enforcement layer without relying on cloud-based inference. Note: deterministic guarantees apply specifically to the rule-enforcement layer (RBSL); upstream perception and VLM components remain probabilistic by design.

4 HYBRID SYSTEM ARCHITECTURE

The ACSAR-E architecture separates probabilistic AI planning from deterministic policy enforcement to ensure verifiable safety on edge hardware.

4.1 Policy-Constrained Decision Loop

The core contribution is a Hybrid Reasoning Engine that acts as a mandatory filter between the VLM and the robot's actuators:

- **VLM Planner:** Generates candidate actions based on real-time environmental context and mission goals.
- **Symbolic Rule Engine:** A deterministic filter that validates every candidate action against compiled institutional policy before execution. Justifications (J) are derived directly from the matched rule/policy path — not generated post hoc by the VLM.
- **Semantic Mapping:** Utilizes RTAB-Map and Fast-LIO2 to annotate 3D zones, augmented by WiFi-derived occupancy heatmaps to identify presence in occluded areas (e.g., behind closed doors) for proactive policy lookup.

Determinism scope and uncertainty behavior: The hybrid architecture provides deterministic guarantees at the enforcement layer only. When scene confidence $c < 0.40$ or sensor inputs conflict (e.g., LiDAR/camera disagreement), the system defaults to a safe hold posture and alerts the operator via TTS, rather than proceeding under uncertainty. Every such event is timestamped in the audit log to support post-hoc compliance review.

4.2 Edge Reliability & Recovery

To maintain "Zero-Cloud" autonomy on NVIDIA Orin NX hardware, the system integrates a Dynamic Recovery Node:

- **Failure Detection:** If the IMU detects a tilt > 60° or a "Fall" state, the PGDS is immediately interrupted.

- **Self-Righting:** The robot executes an automated recovery sequence via the Unitree SDK and re-localizes using the XT16 LiDAR before resuming the patrol mission.
- **VLM Invocation Strategy:** To prevent thermal throttling from affecting the real-time safety path, the VLM pipeline is decoupled from the RBSL control loop. The VLM is invoked selectively on scene-change events or policy-lookup triggers not on every control cycle ensuring VLM latency variation never degrades time-critical safety responses.

5 POLICY-GROUNDED DECISION SCHEMA (PGDS)

5.1 Framework f (S, P) → (A, c, J)

- **Scene Graph (S):** Multi-modal input combining 10 Hz YOLOv8n spatial data with WiFi-sensing trajectory vectors to represent both visible and occluded agents.
- **Policy (P):** Executable YAML constraints.
- **Outputs:** Action (A), Confidence (c), Justification (J). J is derived from the matched rule/policy clause — not post-hoc AI generation.

5.2 Safety & Latency (RBSL)

A 200 Hz C++ gatekeeper ensures:

- **Action Masking:** Overrides if c < 0.40; defaults to safe hold posture.
- **Real-time Guard:** < 0.4 s first-token latency; 3 s emergency stop trigger.

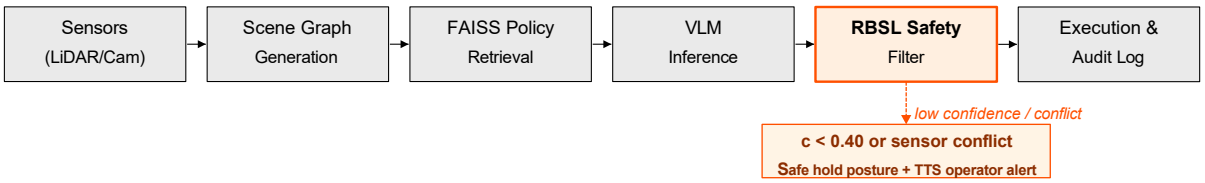

Figure 1: PGDS Operational Logic (Horizontal Flow). Dashed orange branch shows added uncertainty fallback: c < 0.40 or sensor conflict triggers safe hold + TTS alert.

6 D1 Arm Whole-Body Control (WBC) Integration

The integration of the Unitree D1 arm with the Go2 quadruped enables physical policy enforcement through a unified Whole-Body Control (WBC) strategy. This allows for precise manipulation and interaction unavailable to standard wheeled platforms.

6.1 Operational Capabilities

- **Active Manipulation:** Door handle operation with a force-controlled grasp (5-15 N torque limit).
- **Hazard Response:** Arm-deployed physical markers at policy-violation sites.
- **Visual Servoing:** Access card retrieval via D435i point cloud; arm re-plans if lever pose error exceeds 5 mm.

6.2 Whole-Body Control Strategy

To maintain equilibrium during high-torque tasks, the system coordinates the arm and base as a single entity:

Table 2: WBC Parameters for D1 Arm Integration

Feature	Technical Implementation
Base Stability	Transitions to a four-point low-stance before arm activation to maximize the support polygon.
Precision	Nav2 maintains base stationarity within ±1 cm via closed-loop foot placement.
Wrench Compensation	Reaction forces from arm operation are fed back to the locomotion controller to neutralize body sway.
Abort Logic	RBSL monitors joint torques at 200 Hz; immediate stow-and-halt if unexpected resistance is detected.

6.3 Software Stack

The architecture utilizes MoveIt2 over ROS2 interfaced with the Unitree Arm SDK. Arm motion is constrained to stationary approaches, remaining stowed during all patrol motion or when velocity v > 0.3 m/s.

7. Hardware & Sensor Integration

The ACSAR-E architecture utilizes a multi-modal sensor suite integrated via ROS2. The table below details the specific hardware components, their operational roles, and the critical system impacts in the event of component failure.

Table 3: ACSAR-E Hardware Integration Matrix

Component	Core Role (ACSAR-E)	Critical Failure Impact
Unitree L1	360°×90° zone mapping; policy boundary classification.	Zone classification collapse.

Component	Core Role (ACSAR-E)	Critical Failure Impact
Hesai XT16	Glass/structural grounding; dual-fusion via Fast-LIO2 (RMSE < 2 mm).	Specular failure untestable.
RealSense D435i	YOLOv8n detection; RBSL e-stop (< 0.5 m); arm grasp estimation.	Safety gating & manipulation fail.
Unitree D1	6-DOF force control (5–15 N); auto-stow (> 0.3 m/s) via MoveIt2.	Physical enforcement lost.
Jetson Orin NX	On-device PGDS+RBSL; 100 TOPS (INT4); 16 GB RAM.	Privacy risk & high latency.
WiFi Sensing Unit	NLOS presence/movement detection; through-wall occupancy counting.	Blind-spot policy violations in partitioned spaces.
Audio/IMU	TTS justifications; IMU tilt e-stop (> 25° @ 200 Hz).	Comms & safety gate removed.

8. Evaluation Framework

200 - 500 scenarios; Ground truth by 2 annotators.

Metric	Target
Macro-avg F_1 (FN weighted 2×)	> 85%
Critical Policy Violation FN	< 5%
FAISS Retrieval Recall (top-3)	> 95%
Inter-rater Agreement (κ)	> 0.75
HRI SUS Score (N=15)	≥ 68
Arm Grasp Accuracy	≥ 90%
RBSL first-token latency	< 0.4 s
Orin NX sustained temperature	< 85°C
VLM throttle latency delta	< 3 s

9. Compute: Jetson Orin NX

Local-only (GDPR/PDPA); 100 TOPS INT4.

Module	RAM	Proc.
LLaVA-1.5 3B	3.5 GB	GPU
YOLOv8n + Track	0.5 GB	GPU
Nav2 / MoveIt2	3.2 GB	CPU
RBSL C++ (200 Hz)	0.3 GB	RT-C
WiFi Signal Processor (C++)	0.4 GB	CPU
OS / Drivers	2.0 GB	—
Headroom	6.1 GB	—

10. Deployment Platform (Unitree Go2 Edu)

Locomotion: Quadrupedal stability for 0.7 m corridors and thresholds (> 25°); active compliance on contact.

Versatility: YAML-based policies for Office (Privacy), Logistics (Hazmat/PPE), and Healthcare (HIPAA).

Thermal Degradation Modes: To support sustained real-world operation, the system implements three graduated modes: (1) *Normal:* all modules active, VLM invoked on scene-change events; (2) *Reduced* (Orin NX > 78°C) VLM invocation rate halved, non-critical logging suspended; (3) *Safety-only* (> 85°C): VLM suspended, RBSL and Nav2 remain fully active, operator alerted via TTS. Temperature and throttle state are logged continuously as runtime metrics.

Like a car's temperature warning: the robot scales back AI processing when hot but safety rules and navigation stay fully active at all times.

11. Simulation-to-Real Validation Strategy

Digital Twin Development: A high-fidelity digital twin of the deployment environment will be developed using NVIDIA Isaac Sim to enable controlled testing and validation.

Wi-Fi Sensing Validation: An RF ray-tracing extension will be integrated to simulate non-line-of-sight (NLOS) agent movement, enabling rigorous stress testing of policy (P) constraints prior to real-world deployment.

Sim-to-Real Risk Mitigation: A 5-pair matched-scenario methodology will be employed to systematically compare Wi-Fi signal processor performance between the simulated environment (Isaac Sim) and the physical Unitree Go2 Edu platform.

12. Risk Register

Risk	Primary Impact	Mitigation
VLM Accuracy	Misclassification	RBSL safety floor; Staged inference.
Thermal Load	Performance drop	30% duty cycle; INT4 quantization; graduated thermal degradation modes.
LiDAR Fail	Specular/Glass blindness	Dual-LiDAR fusion; Wi-Fi-sensing cross-validation for glass/specular environments; Nav2 cost map inflation.
Arm Collision	Over force (> 15 N)	200 Hz Force monitoring; Auto-stow.
Sim-to-Real	Logic divergence	5-pair matched scenarios; PGDS ≥ 90%.
Data Security	Policy tampering	Ed25519 signing; Schema validation.